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Agile Sprint Planning & User Story Workshop

Project Title : AI-Powered Digital Twin Platform for Smart City Infrastructure

1. Project Vision and Stakeholders

Project Vision

The AI-Powered Digital Twin Platform for Smart City Infrastructure is designed to create a virtual representation of physical city assets such as roads, bridges, buildings, water pipelines, streetlights, transportation systems, and utility networks. By integrating Artificial Intelligence (AI), Internet of Things (IoT), Geographic Information Systems (GIS), and Cloud Computing, the platform continuously monitors infrastructure conditions using real-time sensor data.

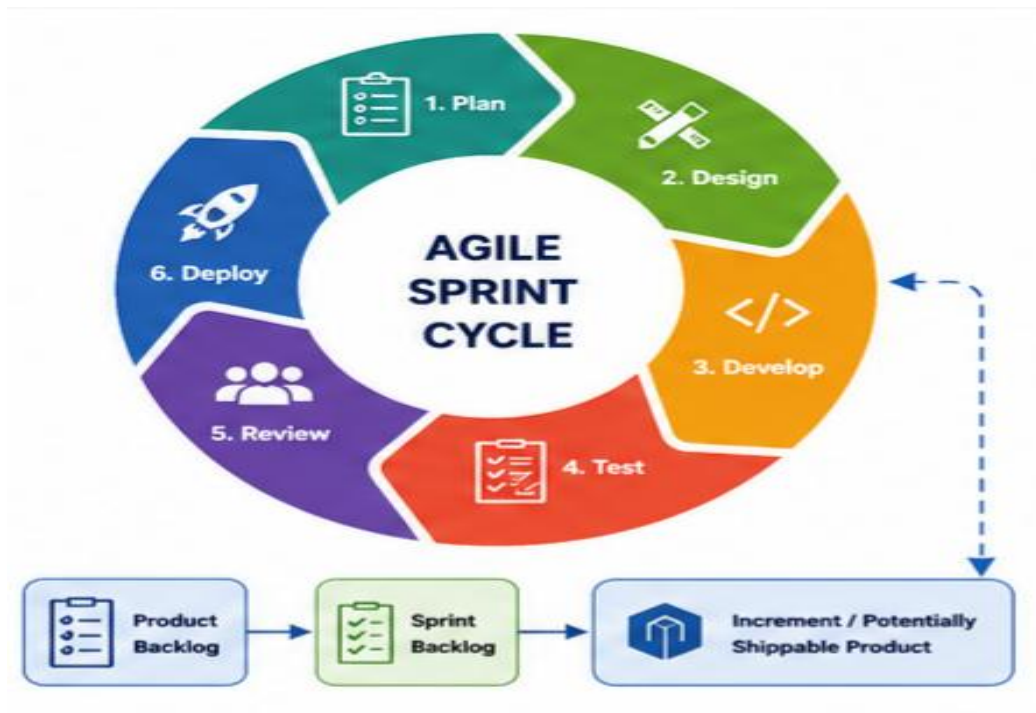
Unlike traditional monitoring systems that only provide historical information, this platform predicts failures before they occur, detects anomalies instantly, and assists authorities in making data-driven maintenance decisions. The digital twin enables engineers and city administrators to simulate different urban scenarios, evaluate maintenance strategies, and improve overall infrastructure planning.

The primary vision of this project is to transform conventional infrastructure management into an intelligent, predictive, and automated system that enhances operational efficiency while reducing maintenance costs.

Objectives

1. Create digital twins of smart city infrastructure.
2. Integrate real-time IoT sensor data.
3. Predict infrastructure failures using AI.
4. Continuously monitor infrastructure health.
5. Support predictive maintenance.

6. Simulate different urban scenarios.
7. Generate infrastructure performance reports.
8. Improve smart city planning and sustainability.



Stakeholders :

Stakeholder	Role
City Administrator	Monitor infrastructure
Maintenance Team	Repair assets
Citizens	Report issues
Government	Planning & funding
IoT Devices	Collect sensor data
AI System	Predict failures
Cloud Platform	Store and process data

2. Epics and User Stories

Epic 1 – Infrastructure Monitoring

User Story 1

As a city administrator, I want to monitor infrastructure health in real time so that I can identify failures quickly.

User Story 2

As a maintenance engineer, I want to receive alerts when abnormal sensor values are detected so that maintenance can begin immediately.

Epic 2 – Digital Twin Management

User Story 3

As a city planner, I want to visualize digital twins of city assets so that infrastructure conditions are easy to understand.

User Story 4

As a system administrator, I want to add or update infrastructure assets so that the digital twin remains accurate.

Epic 3 – AI Failure Prediction

User Story 5

As an AI analyst, I want the system to predict infrastructure failures using machine learning so that preventive maintenance can be scheduled.

User Story 6

As a maintenance manager, I want AI-generated risk scores so that critical assets receive priority.

Epic 4 – Predictive Maintenance

User Story 7

As a maintenance engineer, I want maintenance schedules generated automatically so that equipment downtime is minimized.

User Story 8

As a city administrator, I want maintenance history stored for future analysis.

Epic 5 – Reporting and Analytics

User Story 9

As a government official, I want infrastructure performance reports so that funding decisions can be made effectively.

User Story 10

As a city planner, I want predictive analytics dashboards so that long-term planning becomes easier.

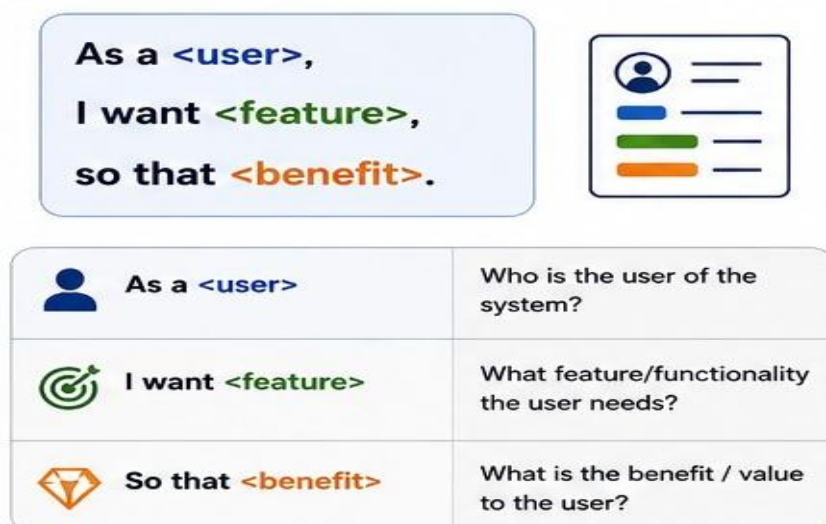
Epic 6 – User Management

User Story 11

As an administrator, I want secure login for authorized users.

User Story 12

As a user, I want role-based access so that sensitive information remains protected.



3. Acceptance Criteria

User Story 1

Given IoT sensors are active

When new sensor data arrives
Then the dashboard updates within 5 seconds.

User Story 2

Given abnormal sensor values occur
When thresholds exceed limits
Then an alert is sent to maintenance staff.

User Story 3

Given infrastructure assets exist
When users open the dashboard
Then the digital twin is displayed correctly.

User Story 4

Given an administrator uploads a new asset
When the information is saved
Then the digital twin updates automatically.

User Story 5

Given sufficient historical data
When AI analyzes infrastructure conditions
Then future failures are predicted with confidence scores.

User Story 6

Given multiple infrastructure assets
When AI calculates risk
Then assets are ranked by priority.

User Story 7

Given predicted failures

When maintenance schedules are generated

Then engineers receive work assignments.

User Story 8

Given maintenance is completed

When engineers update records

Then maintenance history is stored successfully.

User Story 9

Given infrastructure data is available

When reports are generated

Then administrators can download them as PDF.

User Story 10

Given analytics are available

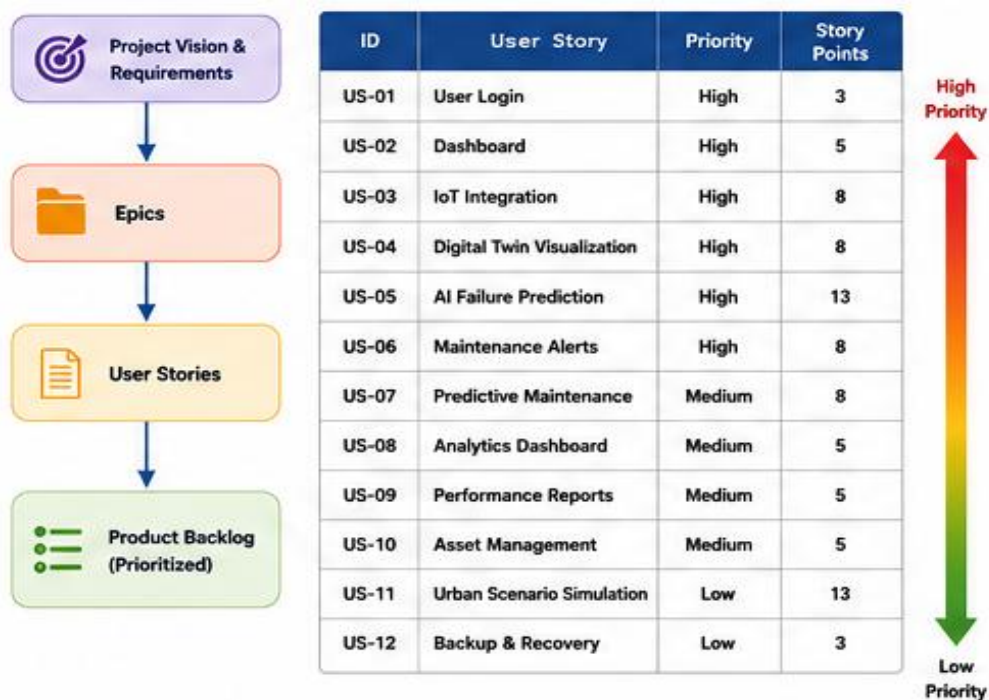
When dashboards are opened

Then graphs and KPIs are displayed correctly.

4. Product Backlog

Priority	Product Backlog Item	Story Points
High	User Login	3
High	Dashboard	5
High	IoT Integration	8
High	Digital Twin Visualization	8

Priority	Product Backlog Item	Story Points
High	AI Failure Prediction	13
High	Maintenance Alerts	8
Medium	Predictive Maintenance	8
Medium	Analytics Dashboard	5
Medium	Performance Reports	5
Medium	Asset Management	5
Low	Urban Scenario Simulation	13



5. Sprint Backlog

Sprint 1 Goal

Develop a working prototype capable of monitoring infrastructure using IoT sensors and displaying a real-time digital twin with AI-based prediction.

Sprint Tasks

Task	Assigned To
Requirement Analysis	Business Analyst
Database Design	Database Engineer

UI Design	Frontend Developer
User Authentication	Backend Developer
Dashboard Development	Frontend Developer
IoT Integration	IoT Engineer
AI Prediction Module	AI Engineer
Notification System	Backend Developer
Testing	QA Engineer
Deployment	DevOps Engineer

6. Story Point Estimation

The Fibonacci estimation technique is used because Agile teams commonly estimate work based on complexity and effort.

User Story	Story Points	Justification
Login	3	Simple user authentication with secure login and basic validation.
Dashboard	5	Requires moderate UI development to display real-time infrastructure status and analytics.
IoT Integration	8	Involves integrating multiple IoT sensors, handling real-time data transmission, and ensuring reliable communication.
Digital Twin Visualization	8	Requires creating an interactive 2D/3D virtual representation of city infrastructure with live updates.
AI Prediction	13	High complexity due to machine learning model development, training, and failure prediction algorithms.

User Story	Story Points	Justification
Maintenance Alerts	8	Implements real-time alert generation using sensor thresholds and AI predictions through notifications.
Reports	5	Generates infrastructure performance reports in PDF/Excel formats with charts and summaries.
Simulation	13	Complex implementation of scenario-based simulations for infrastructure planning and predictive analysis.

7. Sprint Goal and Expected Deliverables

Sprint Goal

The objective of Sprint 1 is to build a functional prototype capable of collecting IoT sensor data, creating digital twins, predicting infrastructure failures, and displaying infrastructure health through an interactive dashboard.

Expected Deliverables

- User Authentication Module
- Infrastructure Dashboard
- Digital Twin Visualization
- IoT Sensor Integration
- AI-Based Failure Prediction
- Maintenance Alert System
- Analytics Dashboard
- Infrastructure Performance Report
- Database Integration
- Cloud Deployment Prototype

8. Conclusion

The AI-Powered Digital Twin Platform for Smart City Infrastructure provides an intelligent solution for monitoring, managing, and maintaining urban infrastructure. Through the integration of AI, IoT, GIS, and cloud technologies, the platform enables real-time visualization, predictive maintenance, and data-driven decision-making.

The Agile Sprint Planning approach ensures that development is incremental, flexible, and customer-focused. Clearly defined epics, user stories, acceptance criteria, backlog prioritization, and sprint planning allow the development team to deliver valuable software efficiently.

The implementation of this platform will reduce maintenance costs, improve infrastructure reliability, enhance operational efficiency, support sustainable city development, and enable smarter urban planning for future smart cities.